Nagar Yuwak Shikshan Sanstha's
Yeshwantrao Chavan College of Engineering
(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)
Hingna Road, Wanadongri, Nagpur - 441 110
NAAC A++
Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu

Department of Computer Technology

Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skills through emerging technologies.

Session 2026-2027

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords:

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Krishna Mishra
Name and Signature of Student and Date
(Signature and Date in Handwritten)

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Session	**2026-27 (ODD)**	**Course Name**	**Lab: Generative AI**
Semester	**7**	**Course Code**	**23ADS1706**
Roll No	**41**	**Name of Student**	**Krishna Mishra**

Practical Number	**7**
Course Outcome	CO1: Explain the fundamentals of Artificial Intelligence, Neural Networks, Natural Language Processing, and Transfer Learning used in Generative AI applications. CO2: Apply Generative AI models such as GANs, VAEs, Transformers, BERT, GPT, and T5to solve real-world AI and NLP problems. CO3: Analyze and optimize prompts using prompt engineering techniques for improving the performance of Large Language Models in different application domains. CO4: Design and develop Generative AI solutions by integrating Foundation Models, Retrieval-Augmented Generation (RAG), tokenization, fine-tuning, and chunking techniques for domain-specific applications.
Aim	To design a question answering system that answers student queries from college regulations/FAQ text using a pre-trained transformer model.
Problem Definition	Design an intelligent assistant capable of answering student queries from college regulations or FAQs using a pre-trained language model.
Theory (100 words)	Question Answering (QA) systems find the answer to a question from a given passage. In extractive QA, a BERT-based model takes the question and the context together as input and predicts the start and end positions of the answer span inside the context. DistilBERT, a smaller version of BERT, fine-tuned on the SQuAD dataset is used here. The model produces a start score and an end score for every token; the tokens between the highest start and end positions are decoded as the answer. This allows students to ask questions in natural language about rules, fees or attendance.
Procedure	Algorithm: • Start.

and Execution (100 Words)	• Load the pre-trained DistilBERT question-answering model and tokenizer. • Store the college regulations/FAQ text as the context. • Take the student's question as input. • Tokenize the question and the context together. • Pass the tokens to the model to get start and end scores. • Select the tokens between the best start and end positions. • Decode the tokens and display the answer. • Repeat for all questions and stop. Code: import torch from transformers import AutoTokenizer, AutoModelForQuestionAnswering model_name = "distilbert-base-cased-distilled-squad" tokenizer = AutoTokenizer.from_pretrained(model_name) model = AutoModelForQuestionAnswering.from_pretrained(model_name) model.eval() context = """ A student must have a minimum attendance of 75 percent in each subject to be eligible for the end semester examination. Students with attendance between 65 and 75 percent may apply for condonation with a medical certificate. The internal assessment carries 40 marks and the end semester examination carries 60 marks. A student must score at least 40 percent overall to pass a course. The library remains open from 9 AM to 6 PM on all working days. Fee payment must be completed before the last date announced by the accounts section, and a late fee of 500 rupees is charged after the due date. Scholarship applications are submitted through the college office within the first month of the semester. """ def answer(question): inputs = tokenizer(question, context, return_tensors="pt", truncation=True, max_length=384) with torch.no_grad(): outputs = model(**inputs)

```
    start = int(outputs.start_logits.argmax())
    end = int(outputs.end_logits.argmax()) + 1

    answer_ids = inputs["input_ids"][0][start:end]

    return tokenizer.decode(
        answer_ids,
        skip_special_tokens=True
    )


questions = [
    "What is the minimum attendance required for the exam?",
    "How many marks does the internal assessment carry?",
    "What is the late fee after the due date?",
    "When is the library open?"
]

for question in questions:
    print("Q:", question)
    print("A:", answer(question))
  print()
```

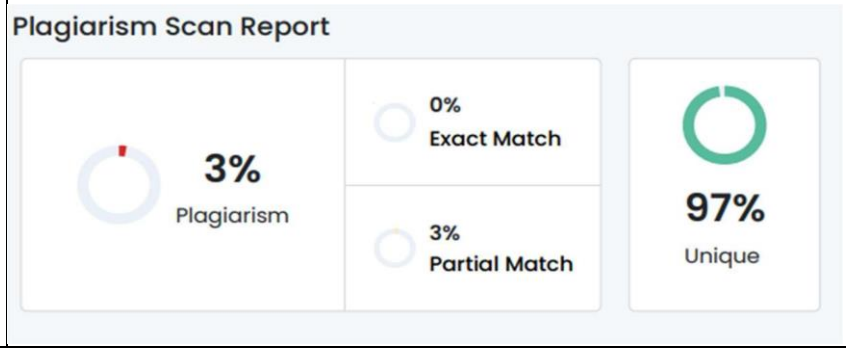
Output Analysis

- Context:
  A sample paragraph of college regulations covering attendance, marks, library timing and fees.
- Questions and Answers:
  The model is expected to extract the following answers from the context:
  What is the minimum attendance required for the exam? → 75 percent
  How many marks does the internal assessment carry? → 40 marks
  What is the late fee after the due date? → 500 rupees
  When is the library open? → 9 AM to 6 PM
- Each answer is a span copied from the context, so the system answers only from the given regulations. Record the exact answers from your run here.

Link of student Github profile where lab assignment has been uploaded.

Github.com/krish-na45/GenAI-Practicals

Plag Report (Similarity index < 12%)



Conclusion	The experiment developed a question answering assistant using a pre-trained DistilBERT model. The model located the answers to student queries inside the college regulations text without any additional training. The system can be extended by using the complete college rule book and a retrieval step to select the relevant section.
Date	/ /2026